AASD 4016
Full Stack Data Science Systems

Applied AI Solutions Developer Program

Module 4
Docker

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Agenda

- Docker
- Why Docker
- Virtual Environment
- Dockerfile
- .dockerignore
- Build a Docker image
- Run a Docker container
- Examples

Iris Classifier

Notebook to Script

Script to Flask

Load model and keep it up for inference

Infer from the model

Test the endpoint using Curl, POSTMAN

Build a python web client

Test the client

Create a Virtual Environment
Install all libraries
Create requirements.txt
Create a Dockerfile
Create a Docker image
Create a Docker container
Test the endpoints within the running Docker container

Current Flask project structure

Project Structure

```
app.py
helper(s).py [Optional]
templates/
    index.html
    result.html
static/
    Style.css [Optional]
```

Expected Flask project structure

Project Structure

```
app.py
helper(s).py [Optional]
templates/
    index.html
    result.html
static/
    Style.css [Optional]
Dockerfile
requirements.txt
iris-venv/
```

Docker

Dockerizing a web application

Installing Docker

Creating a Virtual environment

Installing necessary libraries

Creating requirements.txt

Creating a Dockerfile

Building a Docker image

Running a Docker container

Testing the endpoints

Virtual Environment

Create a new virtual environment

```
virtualenv iris-venv -p python3.8
```

Activate the newly created virtual environment

```
source iris-venv/bin/activate
```

```
deactivate
```

Installing necessary libraries

Install all libraries

```
pip install pandas sklearn Flask
```

Creating requirements.txt

```
pip freeze -l > requirements.txt
```

requirements.txt

Creating requirements.txt

```
pip freeze -l > requirements.txt
```

```
click==8.0.1
Flask==2.0.1
itsdangerous==2.0.1
Jinja2==3.0.1
joblib==1.0.1
MarkupSafe==2.0.1
numpy==1.21.1
pandas==1.3.1
python-dateutil==2.8.2
pytz==2021.1
scikit-learn==0.24.2
scipy==1.7.1
six==1.16.0
sklearn==0.0
threadpoolctl==2.2.0
Werkzeug==2.0.1
```

Dockerfile

```
FROM python:3.8.5

ADD . /app

WORKDIR /app

RUN pip install -r requirements.txt

ENTRYPOINT ["python"]

CMD ["app.py"]
```

Build docker image

Building a Docker image

```
docker build -t iris:1.0 .
```

Sending build context to Docker daemon **345.6MB**

Step 1/6 : From python:3.8

---> 42d342fa9773

Build docker image

```
Sending build context to Docker daemon   345.6MB
Step 1/6 : From python:3.8
---> 42d342fa9773
Step 2/6 : ADD . /app
---> 71124d1c0444
Step 3/6 : WORKDIR /app
---> Running in cbb0584cb001
Removing intermediate  container cbb0584cb001
---> 7612619ec6ab
Step 4/6 : RUN pip install -r requirements.txt
---> Running in 1f8d962449d7
Collecting click==8.0.1
  Downloading click-8.0.1-py3-none-any.whl (97 kB)
Collecting Flask==2.0.1
  Downloading Flask-2.0.1-py3-none-any.whl (94 kB)
Collecting itsdangerous==2.0.1
  Downloading itsdangerous-2.0.1-py3-none-any.whl (18 kB)
Collecting Jinja2==3.0.1
  Downloading Jinja2-3.0.1-py3-none-any.whl (133 kB)
Collecting joblib==1.0.1
  Downloading joblib-1.0.1-py3-none-any.whl (303 kB)
Collecting MarkupSafe==2.0.1
  Downloading MarkupSafe-2.0.1-cp38-cp38-manylinux2010_x86_64.whl (30 kB)
Collecting numpy==1.21.1
  Downloading numpy-1.21.1-cp38-cp38-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (15.8 MB)
Collecting pandas==1.3.1
  Downloading pandas-1.3.1-cp38-cp38-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (11.7 MB)
Collecting python-dateutil==2.8.2
  Downloading python_dateutil-2.8.2-py2.py3-none-any.whl (247 kB)
Collecting pytz==2021.1
  Downloading pytz-2021.1-py2.py3-none-any.whl (510 kB)
Collecting scikit-learn==0.24.2
  Downloading scikit_learn-0.24.2-cp38-cp38-manylinux2010_x86_64.whl (24.9 MB)
Collecting scipy==1.7.1
  Downloading scipy-1.7.1-cp38-cp38-manylinux_2_5_x86_64.manylinux1_x86_64.whl (28.4 MB)
Collecting six==1.16.0
  Downloading six-1.16.0-py2.py3-none-any.whl (11 kB)
Collecting sklearn==0.0
  Downloading sklearn-0.0.tar.gz  (1.1 kB)
Collecting threadpoolctl==2.2.0
  Downloading threadpoolctl-2.2.0-py3-none-any.whl  (12 kB)
Collecting Werkzeug==2.0.1
  Downloading Werkzeug-2.0.1-py3-none-any.whl  (288 kB)
Building wheels for collected packages: sklearn
  Building wheel for sklearn (setup.py): started
  Building wheel for sklearn (setup.py): finished with status 'done'
  Created wheel for sklearn: filename=sklearn-0.0-py2.py3-none-any.whl   size=1316 sha256=d8493593d1247e1a8e514ef41b0a87e79cbe47477c88397406af13760e784424
  Stored in directory: /root/.cache/pip/wh eels/22/0b/40/fd3f795caaa1fb4c6cb738bc1f56100be1e57da95849bfc897
Successfully built sklearn
Installing collected packages: numpy, threadpoolctl, six, scipy, MarkupSafe, joblib, Werkzeug, scikit-learn, pytz, python-dateutil, Jinja2, itsdangerous, click, sklearn, pandas, Flask
Successfully installed Flask-2.0.1 Jinja2-3.0.1 MarkupSafe-2.0.1 Werkzeug-2.0.1 click-8.0.1 itsdangerous-2.0.1 joblib-1.0.1 numpy-1.21.1 pandas-1.3.1 python-dateutil-2.8.2 pytz-2021.1 scikit-learn-0.24.2 scipy-1.7.1 six-1.16.0 sklearn-0.0 threadpoolctl-2.2.0
Removing intermediate  container c05766dfee41
---> d73b0c5557eb
Step 5/6 : ENTRYPOINT ["python"]
---> Running in 720e69ce7d1b
Removing intermediate  container 720e69ce7d1b
---> 4ea3ba2257b2
Step 6/6 : CMD ["app.py"]
---> Running in 3771137b8938
Removing intermediate  container 3771137b8938
---> def36c7facef
Successfully built def36c7facef
Successfully tagged iris:1.0
```

View docker image

Viewing Docker images in your machine

```
docker images
```

```
(iris-venv) (base) Subashs-MacBook-Pro:L4_Docker subashgandyer$ docker images
REPOSITORY                          TAG       IMAGE ID       CREATED               SIZE
iris                                1.0       def36c7facef   About a minute ago    1.63GB
vqa                                 7.0       35dfc43290d6   2 weeks ago           1.42GB
gcr.io/test-ch-voice-mvp/vqa        7.0       35dfc43290d6   2 weeks ago           1.42GB
vqa                                 6.0       e448840dd4f1   2 weeks ago           1.42GB
gcr.io/test-ch-voice-mvp/vqa        6.0       e448840dd4f1   2 weeks ago           1.42GB
voiceqa-deploy_flask                latest    ec8c6a29a77f   2 weeks ago           1.44GB
voiceqa-deploy_celery               latest    fcfb043cc3f2   2 weeks ago           2.07GB
voiceqa-deploy-test1_web            latest    2c7c5d33c17f   3 weeks ago           1.43GB
voiceqa-deploy-test1_worker_1       latest    2c7c5d33c17f   3 weeks ago           1.43GB
vqa-deploy-web                      latest    9d14573f9335   3 weeks ago           1.44GB
```

dockerignore

.dockerignore

```
iris-venv
.vscode
```

Build another docker image

Building a Docker image

```
docker build -t iris:2.0 .
```

Sending build context to Docker daemon **48.13kB**

Step 1/6 : From python:3.8

---> 42d342fa9773

```
(iris-venv) (base) Subashs-MacBook-Pro:L4_Docker subashgandyer$ docker images
REPOSITORY                          TAG       IMAGE ID       CREATED              SIZE
iris                                2.0       eb9f6bc9ed6b   About a minute ago   1.29GB
iris                                1.0       def36c7facef   5 minutes ago        1.63GB
vqa                                 7.0       35dfc43290d6   2 weeks ago          1.42GB
gcr.io/test-ch-voice-mvp/vqa        7.0       35dfc43290d6   2 weeks ago          1.42GB
vqa                                 6.0       e448840dd4f1   2 weeks ago          1.42GB
gcr.io/test-ch-voice-mvp/vqa        6.0       e448840dd4f1   2 weeks ago          1.42GB
voiceqa-deploy_flask                latest    ec8c6a29a77f   2 weeks ago          1.44GB
voiceqa-deploy_celery               latest    fcfb043cc3f2   2 weeks ago          2.07GB
voiceqa-deploy-test1_web            latest    2c7c5d33c17f   3 weeks ago          1.43GB
voiceqa-deploy-test1_worker_1       latest    2c7c5d33c17f   3 weeks ago          1.43GB
vqa-deploy-web                      latest    9d14573f9335   3 weeks ago          1.44GB
```

Slimmer Dockerfile

```
FROM python:3.8-slim
ADD . /app
WORKDIR /app
RUN pip install -r requirements.txt
ENTRYPOINT ["python"]
CMD ["app.py"]
```

Build slimmer docker image

Building a Docker image

```
docker build -t iris:3.0 .
```

Sending build context to Docker daemon **48.13kB**

Step 1/6 : From python:3.8-slim

---> 42d342fa9773

```
iris    3.0    dd2d53d2f00a    About a minute ago    524MB
```

Run the slimmer docker image

Running a Docker image

```
docker run --name iris -p 5000:5000 iris:3.0
```

```
* Serving Flask app 'app' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: on
* Running on all addresses.
  WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://172.17.0.3:5000/ (Press CTRL+C to quit)
* Restarting with stat
* Debugger is active!
* Debugger PIN: 251-657-431
```

View docker containers

Viewing Docker containers in your machine

```
docker ps
```

```
(iris-venv) (base) Subashs-MacBook-Pro:L4_Docker subashgandyer$ docker ps
CONTAINER ID        IMAGE                      COMMAND                    CREATED              STATUS              PORTS                      NAMES
4c90ab13f4ba        iris:3.0                   "python app.py"            About a minute ago   Up About a minute   0.0.0.0:5000->5000/tcp     iris
2fae7112890d        redis                      "docker-entrypoint.s…"     2 weeks ago          Up 40 hours         0.0.0.0:6379->6379/tcp     beautiful
_cray
ff97afcce927        voiceqa-deploy-test1_web   "python ./app.py"          3 weeks ago          Up 40 hours         0.0.0.0:5002->5002/tcp     voiceqa-d
eploy-test1_web_1
```

Testing the docker container endpoints

Curl

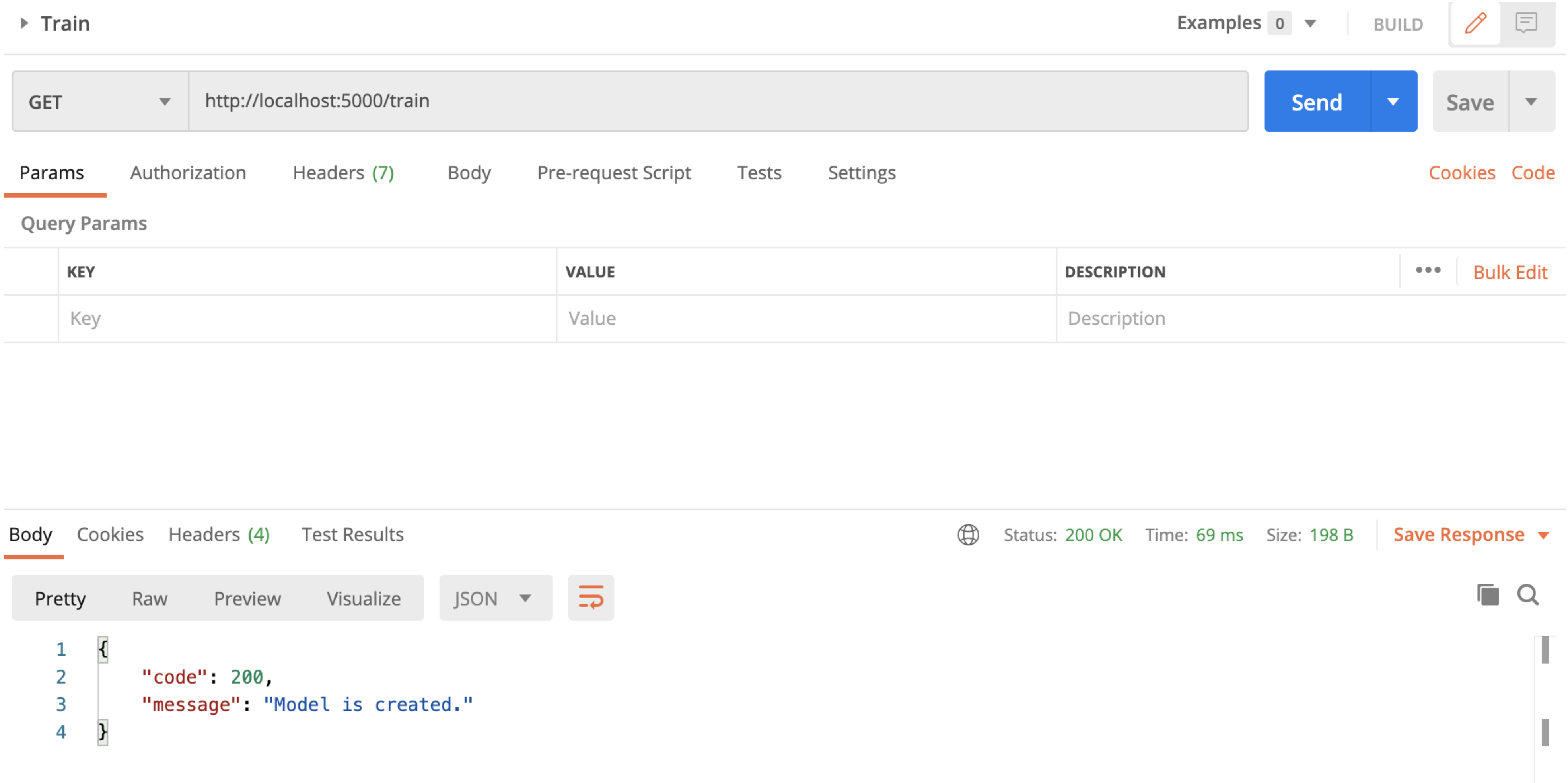
POSTMAN

Python Web Client in browser

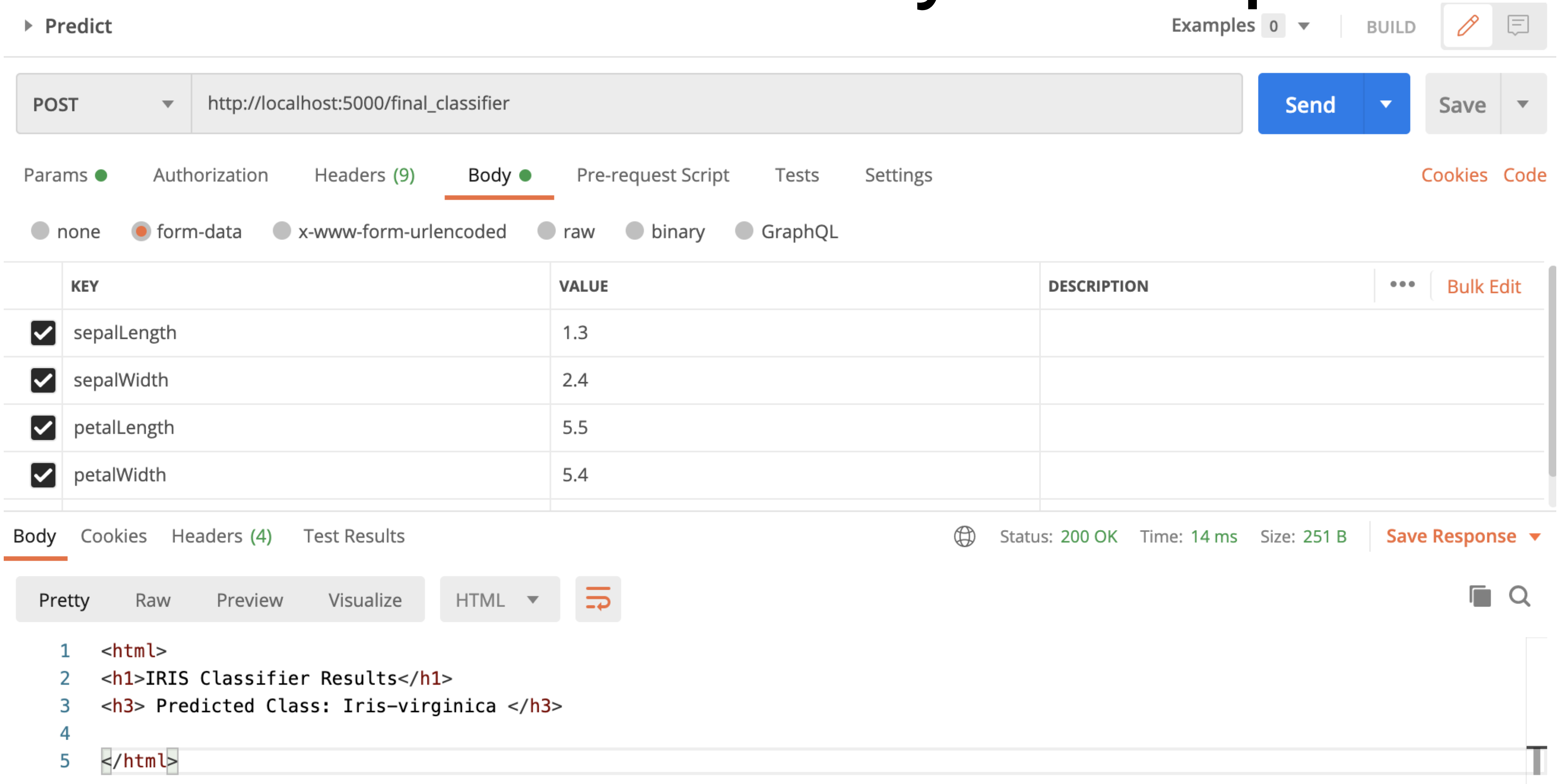
Curl Calls

```
curl http://localhost:5000/train

curl http://localhost:5000/

curl http://localhost:5000/final_classifier?sepalLength=5.1&sepalWidth=3.2&petalLength=3.5&petalWidth=4.2
```

POSTMAN – train a model



POSTMAN – classify a sample



Test with Python Web Client

Naviagate to

    http://localhost:5000/train

Then,

    http://localhost:5000/

Input the values within all fields

Click Classify button

Further Reading

Flask Tutorial

https://flask.palletsprojects.com/en/2.0.x/tutorial/

Docker Tutorial

https://www.docker.com